

# Customer Support Ticketing System (SaaS Helpdesk)

Comprehensive Pin-to-Pin Evaluation Manual for Team Members, Evaluators, and Faculty

**Quick Architecture Summary:** A full-stack, 4-tier Monorepo combining React 19 Frontend, Java 17 Spring Boot 3.3 Backend, Python 3.11 FastAPI NLP Classifier (TF-IDF + Logistic Regression), and PostgreSQL 16 (14 BCNF Tables) with Graph-based BFS Escalation Routing.

## 1. 👥 User Roles & Demo Credentials

All demo accounts share the password: Password123!

| Role                 | Account Email                  | Team / Scope         | Escalation Node    | Key Privileges                                       |
|----------------------|--------------------------------|----------------------|--------------------|------------------------------------------------------|
| Customer             | john.doe@acme.com              | Acme Corporation     | Client             | Submit tickets, view own history, customer replies   |
| Agent (Tech L1)      | elena.rodriguez@supportdesk.io | Technical Support    | L1_SUPPORT         | Initial diagnostics, agent replies, internal notes   |
| Agent (Billing Lead) | sarah.chen@supportdesk.io      | Billing & Finance    | BILLING_SPECIALIST | Financial triage, invoice corrections, refunds       |
| Agent (Tech Lead)    | lucas.muller@supportdesk.io    | Technical Support    | TECHNICAL_TEAM     | Code defect triage, patch reviews                    |
| Agent (Senior Arch)  | david.kim@supportdesk.io       | Technical Support    | SENIOR_ENGINEER    | Infrastructure faults, kernel & scale failures       |
| Admin                | admin@supportdesk.io           | Executive Operations | EXECUTIVE_LEAD     | Global system oversight, team directory, audit trail |

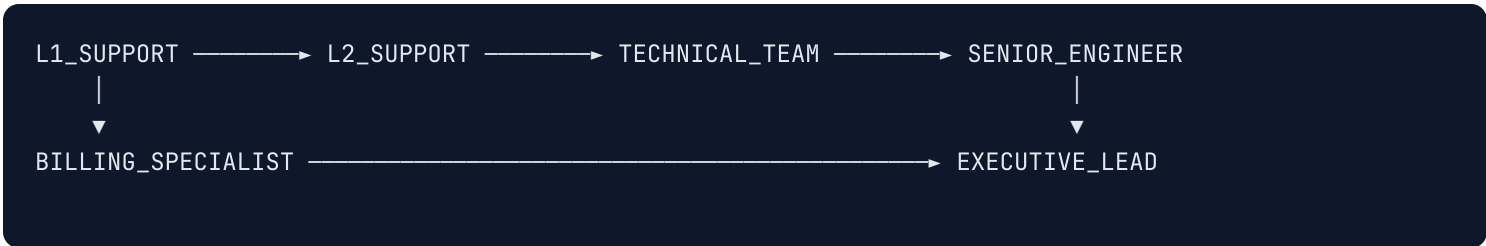
## 2. 📊 Dashboard & Mobile KPI Carousel

The operational center of the helpdesk provides real-time visibility into tickets and SLA compliance.

- **6 KPI Cards (Mobile Swipe Carousel):** Total Tickets, Open Queue, In Progress, Escalated, Resolved, and Average SLA Response ( 4.2h vs < 8.0h target). On mobile, swipes horizontally with 6 active slide indicator dots.
- **Ticket Status & Lifecycle Donut Graph:** Multi-color SVG ring chart showing proportions of Open, In Progress, Waiting Customer, Escalated, Resolved, and Closed tickets with dynamic center stats.
- **1-Click Status Filtering:** Tapping any status bar or donut slice filters the tickets table instantly.
- **Mobile Navigation Drawer:** Fixed 256px sidebar slides in cleanly on mobile via the top hamburger button ( ≡ ) with background blur.

## 3. 🔄 Escalation Graph Architecture (ADSA Graph Theory)

Ticket escalation is modeled as a **Directed Acyclic Graph (DAG)**:  $G = (V, E, W)$  using an in-memory **Adjacency List** ( $\mathcal{O}(|V| + |E|) = 12$  references).



- **BFS Shortest Path Routing (  $\mathcal{O}(|V| + |E|)$  ):** Computes the minimum-hop route from current tier to destination tier.
- **DFS Tri-Color Cycle Prevention:** Confirms no backward cycles exist ( WHITE , GRAY , BLACK states), guaranteeing the network is a strict DAG.
- **Reviewer Scenarios:**
  - *Scenario 1 (Tech Outage):* L1 → L2 → Tech → Senior (3 hops, weight=6, Agent: David Kim).
  - *Scenario 2 (Billing Dispute):* L1 → Billing → Executive (2 hops, weight=3, Agent: Alex Morgan).

## 4. 🗨️ NLP Classifier Lab (Machine Learning)

Automated ticket categorization microservice running Python 3.11 and FastAPI:

- **Text Vectorization:** Uses **TF-IDF** (Term Frequency–Inverse Document Frequency) with unigram and bigram n-grams.
- **Classification Algorithm:** **Multinomial Logistic Regression** with L2 regularization.
- **7 Target Categories:** `BILLING` , `TECHNICAL_SUPPORT` , `ACCOUNT_ACCESS` , `SHIPPING` , `REFUND` , `PRODUCT_ISSUE` , and `GENERAL_INQUIRY` .
- **Interactive Sandbox:** Reviewers can type any message in the *NLP Classifier Lab* tab and see real-time probability bars and automated team assignment.

## 5. 📐 Relational Algebra Query Engine (DBMS & DMGT)

Academic visualizer illustrating mathematical querying over PostgreSQL:

- **$\sigma$  (Selection):** Row filtering (  `$\sigma_{\{status='OPEN'\}}(TICKETS)$`  ).
- **$\pi$  (Projection):** Column selection (  `$\pi_{\{ticket\_number, subject\}}(TICKETS)$`  ).
- **$\bowtie$  (Natural Join):** Relational joining (  `$TICKETS \bowtie_{\{customer\_id\}} CUSTOMERS$`  ).
- **$\cup, \cap, -$  (Set Operations):** Union, Intersection, and Set Difference.
- Side-by-side view showing the mathematical formula, Tuple Relational Calculus (TRC), and live generated SQL query.

## 6. 📁 Monorepo Codebase Browser

An in-app code inspector allowing reviewers to review production code directly in the browser:

- **Backend:** Java 17 Spring Boot (Security, Routing, Graph BFS algorithms).
- **Classifier:** Python 3.11 FastAPI (Model training, predictor, endpoints).
- **Database:** PostgreSQL 16 (14 BCNF tables, foreign keys, triggers, audit schema).

## 7. 🧪 Official 9–Step Reviewer Evaluation Walkthrough

Step-by-step evaluation flow for faculty, judges, and reviewers:

### 1 Login as Customer

Use the top-right role switcher to select **Customer: John Doe (Acme Corp)**.

## 2 Create Test Ticket

Click "Create New Ticket", then click the preset: *"Payment deducted twice"*.

## 3 Verify Python ML Classification

Observe the AI predict Category = `BILLING` with ~ `96.2%` confidence.

## 4 Verify Automated Java Routing

Click Submit. The backend assigns the ticket to **Billing & Finance Team** (Sarah Chen).

## 5 Switch Role to Agent: Sarah Chen

Switch role to Sarah Chen. Open the newly created ticket in the queue.

## 6 Agent Response & Internal Note

Add a customer reply and a private internal note. Observe status change to `IN_PROGRESS`.

## 7 Escalate Ticket via Graph BFS

Click "Escalate Ticket" → choose destination `SENIOR_ENGINEER`. The graph computes the BFS route: `L1 → L2 → Tech → Senior`.

## 8 Verify Escalation Audit Record

The escalation is committed to PostgreSQL `ticket_escalations` table, and status becomes `ESCALATED`.

## 9 Admin Dashboard & Analytics

Switch to **Admin: Alex Morgan** to view the real-time Status Donut Chart, Escalation Graph Explorer, and Relational Algebra Query Engine.